

Test Case Scenarios

Scenario A: Basic Mixed Workload

A normal workload with different arrival times and burst times to verify basic correctness of both algorithms.

Input Data

PID	Arrival Time	Burst Time	Priority
P1	0	8	3
P2	1	4	1
P3	2	9	4
P4	3	5	2
P5	4	2	5

SJF Results

Input Panel

Process ID: Arrival: Burst: Priority (Lower=Higher): Add Process Remove Selected Restart All

Basic Mixed Workload

ID	Arrival	Burst	Priority
P1	0	8	3
P2	1	4	1
P3	2	9	4
P4	3	5	2
P5	4	2	5

Run Simulation & Compare

SJF Results

Priority Results

Comparison & Conclusion

SJF Gantt Chart

P1

P2

P5

P4

P1

P3

0157121928

ID	Arrival	Burst	Priority	CT	WT	TAT	RT
P1	0	8	3	19	11	19	8
P2	1	4	1	5	0	4	0
P3	2	9	4	28	17	26	17
P4	3	5	2	12	4	9	4
P5	4	2	5	7	1	3	1

PID	Arrival	Burst	CT	WT	TAT	RT
P1	0	8	19	11	19	0
P2	1	4	5	0	4	0
P3	2	9	28	17	26	17
P4	3	5	12	4	9	4
P5	4	2	7	1	3	1

Avg WT: 6.6 | Avg TAT: 12.2 | Avg RT: 4.4

Priority Scheduling Results

Input Panel

Process ID: Arrival: Burst: Priority (Lower=Higher): Add Process Remove Selected Restart All Basic Mixed WorkLo

ID	Arrival	Burst	Priority
P1	0	8	3
P2	1	4	1
P3	2	9	4
P4	3	5	2
P5	4	2	5

Run Simulation & Compare

SJF Results Priority Results Comparison & Conclusion

Priority Gantt Chart

ID	Arrival	Burst	Priority	CT	WT	TAT	RT
P1	0	8	3	17	9	17	0
P2	1	4	1	5	0	4	0
P3	2	9	4	26	15	24	15
P4	3	5	2	10	2	7	2
P5	4	2	5	28	22	24	22

PID	Arrival	Burst	CT	WT	TAT	RT
P1	0	8	17	9	17	0
P2	1	4	5	0	4	0
P3	2	9	26	15	24	15
P4	3	5	10	2	7	2
P5	4	2	28	22	24	22

Avg WT: 9.6 | Avg TAT: 15.2 | Avg RT: 7.8

Analysis

- *Waiting Time: SJF achieved a lower average waiting time (6.6 vs 9.6).*
- *Turnaround: SJF achieved a lower average turnaround time (12.2 vs 15.2).*
- *Response Time: SJF gave processes CPU access sooner (4.4 vs 7.8).*

- *Fairness: SJF was fairer - its worst-case wait was 17 vs 22.*
- *Starvation Risk: A large gap between max and average WT suggests possible starvation for at least one process.*

- *Recommendation: For this workload, SJF is the better choice overall.
SJF minimizes wait for short-burst jobs but may starve long ones.
Priority favors urgent tasks but may delay low-priority work.*

Scenario B: Conflict Between Burst Time and Priority

A workload designed to reveal a meaningful difference: a short-burst process has low priority, while a long-burst process has high priority.

Input Data

PID	Arrival Time	Burst Time	Priority
P1	0	2	5
P2	0	10	1
P3	1	3	4
P4	2	1	3

SJF Results

Input Panel

Process ID: Arrival: Burst: Priority (Lower=Higher): Add Process Remove Selected Restart All Conflict between B

ID	Arrival	Burst	Priority
P1	0	2	5
P2	0	10	1
P3	1	3	4
P4	2	1	3

Run Simulation & Compare

SJF Results

Priority Results

Comparison & Conclusion

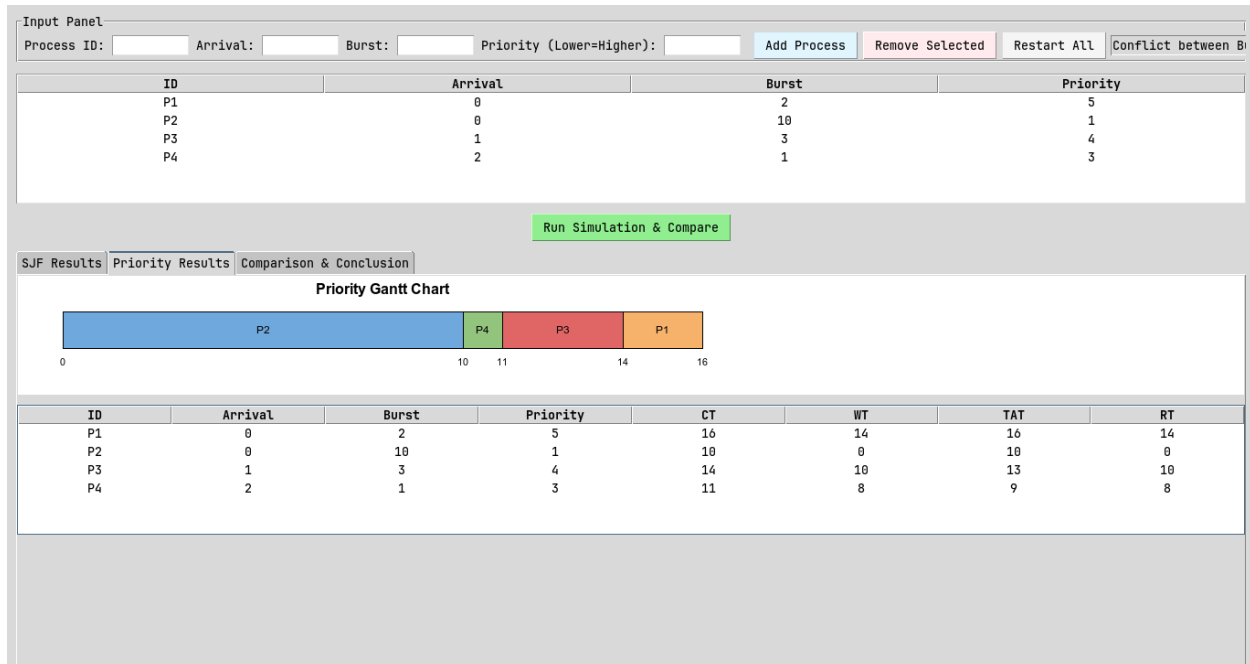
SJF Gantt Chart

ID	Arrival	Burst	Priority	CT	WT	TAT	RT
P1	0	2	5	2	0	2	0
P2	0	10	1	16	6	16	6
P3	1	3	4	6	2	5	2
P4	2	1	3	3	0	1	0

PID	Arrival	Burst	CT	TAT	WT	RT
P1	0	2	2	2	0	0
P2	0	10	16	16	6	6
P3	1	3	6	5	2	2
P4	2	1	3	1	0	0

Avg WT: 2.0 | Avg TAT: 6.0 | Avg RT: 2.0

Priority Scheduling Results



PID	Arrival	Burst	CT	TAT	WT	RT
P1	0	2	16	16	14	14
P2	0	10	10	10	0	0
P3	1	3	14	13	10	10
P4	2	1	11	9	8	8

Avg WT: 8.0 | Avg TAT: 12.0 | Avg RT: 8.0

Analysis

- **Waiting Time:** SJF achieved a lower average waiting time (2.0 vs 8.0).
- **Turnaround:** SJF achieved a lower average turnaround time (6.0 vs 12.0).
- **Response Time:** SJF gave processes CPU access sooner (2.0 vs 8.0).
- **Fairness:** SJF was fairer - its worst-case wait was 6 vs 14.
- **Starvation Risk:** A large gap between max and average WT suggests possible starvation for at least one process.
- **Recommendation:** For this workload, SJF is the better choice overall. SJF minimises wait for short-burst jobs but may starve long ones. Priority favours urgent tasks but may delay low-priority work.

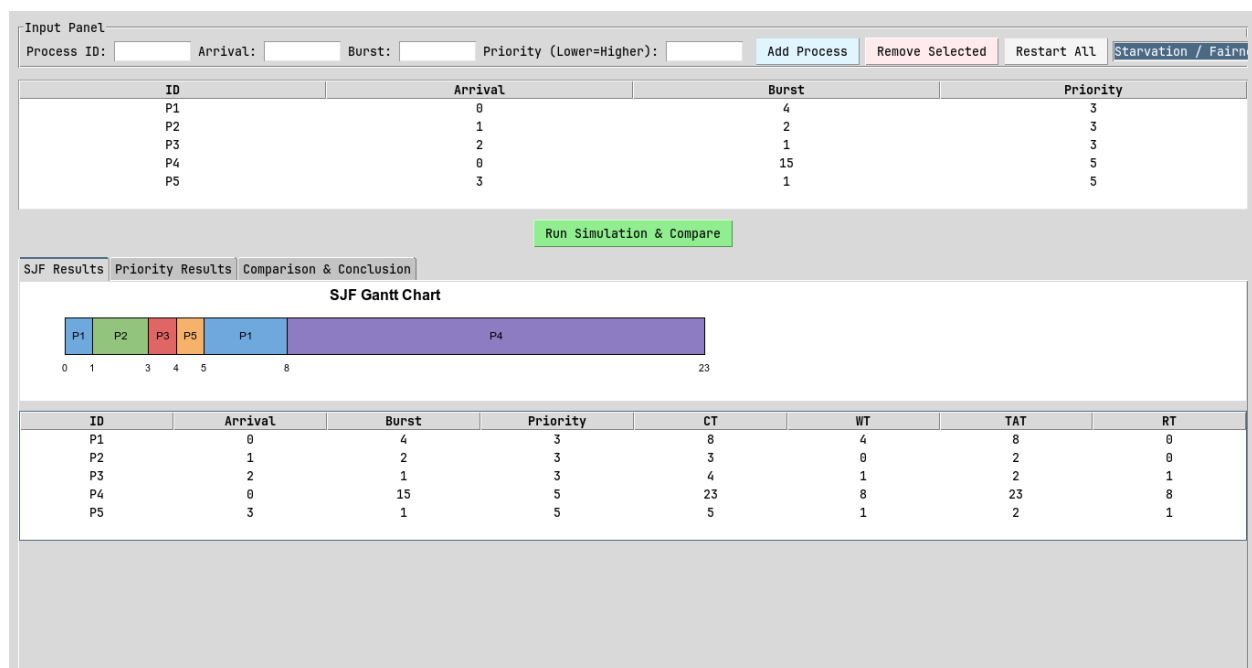
Scenario C: Starvation / Fairness Case

A workload where one process may wait much longer under one of the algorithms, demonstrating unfair delay or starvation.

Input Data

PID	Arrival Time	Burst Time	Priority
P1	0	4	3
P2	1	2	3
P3	2	1	3
P4	0	15	5
P5	3	1	3

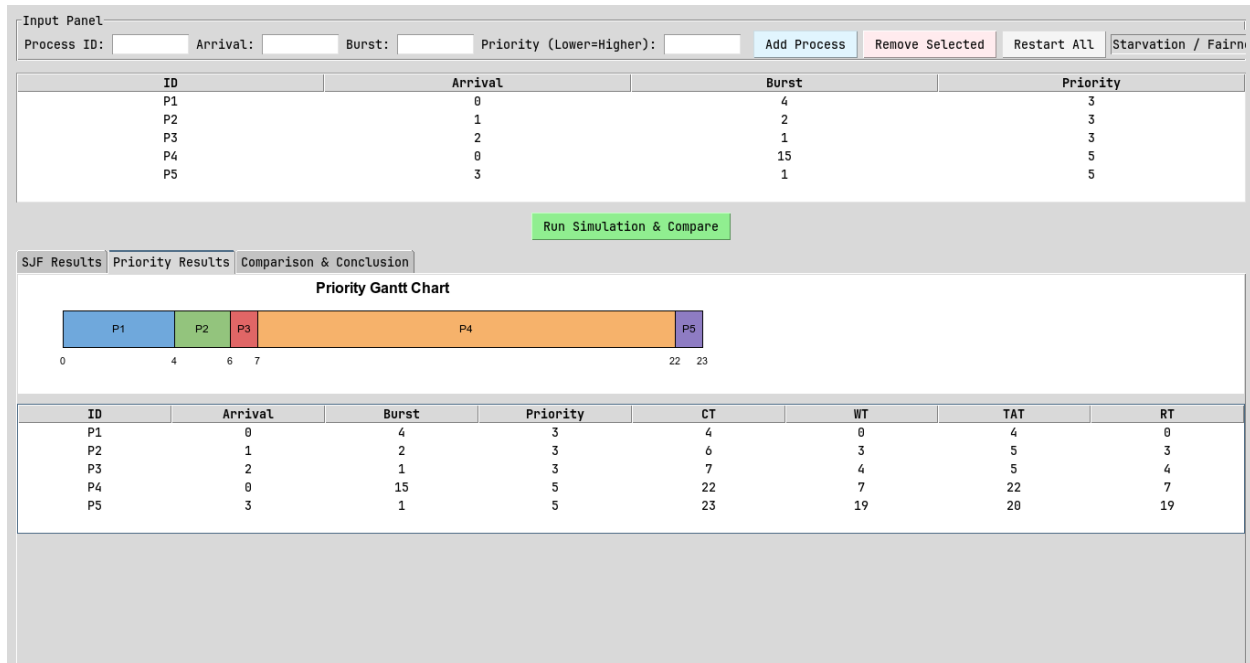
SJF Results



PID	Arrival	Burst	CT	TAT	WT	RT
P1	0	4	8	8	4	0
P2	1	2	3	2	0	0
P3	2	1	4	2	1	1
P4	0	15	23	23	8	8
P5	3	1	5	2	1	1

Avg WT: 2.8 | Avg TAT: 7.4 | Avg RT: 2.0

Priority Scheduling Results



PID	Arrival	Burst	CT	TAT	WT	RT
P1	0	4	4	4	0	0
P2	1	2	6	5	3	3
P3	2	1	7	5	4	4
P4	0	15	22	22	7	7
P5	3	1	23	20	19	19

Avg WT: 6.6 | Avg TAT: 11.2 | Avg RT: 6.6

Analysis

- *Waiting Time:* SJF achieved a lower average waiting time (2.8 vs 6.6).
- *Turnaround:* SJF achieved a lower average turnaround time (7.4 vs 11.2).
- *Response Time:* SJF gave processes CPU access sooner (2.0 vs 6.6).
- *Fairness:* SJF was fairer - its worst-case wait was 8 vs 19.
- *Starvation Risk:* A large gap between max and average WT suggests possible starvation for at least one process.
- *Recommendation:* For this workload, SJF is the better choice overall.
SJF minimises wait for short-burst jobs but may starve long ones.
Priority favours urgent tasks but may delay low-priority work.

Scenario D: Input Validation Case

This scenario demonstrates the simulator's input validation by attempting to submit invalid data. No simulation is run; the purpose is to confirm errors are caught and shown correctly.

Invalid Input Attempted	Expected Behavior
Burst Time = 0	Burst time must be greater than zero.
Arrival Time = -1	Arrival time cannot be negative.
Input field = 'abc'	Arrival, Burst, and Priority must be integers.
Duplicate PID	Duplicate process ID
Empty field left blank	All fields (ID, Arrival, Burst, Priority) are required.
Priority left empty or invalid	Priority cannot be negative.